

Sequences

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1 Arithmetic Sequence

Definition 1.1. *Given a starting value of a_0 and a constant difference value of d the arithmetic sequence is defined as follows.*

$$\boxed{a_n = a_0 + n \cdot d}$$

1.1 Sum of Arithmetic sequence

Theorem 1.1.

$$\sum_{k=0}^n a_k = \frac{(n+1)}{2} (2a_0 + n \cdot d) \quad (1)$$

Proof

Define S_n such that:

$$S_n = \sum_{k=0}^n a_k$$

Note that

$$2S_n = \sum_{k=0}^n a_k + \sum_{k=0}^n a_{n-k} = \sum_{k=0}^n (a_k + a_{n-k})$$

Using the definition we see $a_k + a_{n-k} = (a_0 + k \cdot d) + (a_0 + (n-k) \cdot d) = 2a_0 + n \cdot d$ which does not depend on k

$$\begin{aligned} 2S_n &= \sum_{k=0}^n (2a_0 + n \cdot d) = (n+1)(2a_0 + n \cdot d) \\ S_n &= \frac{(n+1)}{2} (2a_0 + n \cdot d) \end{aligned}$$

1.2 Example: Sum of Natural Numbers

Prove that

$$\sum_{i=0}^n i = 0 + 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Proof: Set $a_0 = 0$ and $d = 1$

$$\sum_{i=0}^n i = \frac{(n+1)}{2}(n)$$